



Relationship between maternal serum sFlt-1 level and placenta accreta spectrum disorders in the third trimester

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Abstract

Purpose This study aims to evaluate whether the third-trimester soluble fms-like tyrosine kinase-1 (sFlt-1) serum levels could be related to placenta accreta spectrum (PAS) disorders and the severity of postpartum blood loss.

Methods This was a nested case–control study which compared serum sFlt-1 level between gravid women with or without PAS disorders. Spearman correlation analysis was conducted to explore the relationship between sFlt-1 level and the volume of postpartum blood loss. Confounding factors were adjusted to avoid the impact on the results.

Results Sixty gravid women were enrolled: 36 women in the PAS group and 24 women in the non-PAS group. Women in the PAS group had a median sFlt-1 level of 9407.1 [2745.9–21,691.5] pg/ml, whereas women in the non-PAS group had a median sFlt-1 level of 25,779.2 [14317.1–35,626.7] pg/ml, ($p < 0.001$). The sFlt-1 level was negatively related to the volume of postpartum blood loss ($r = -0.358$, $p = 0.041$). After adjusting for maternal age and gestational age at blood taking, sFlt-1 level showed no significant relationship with PAS disorders ($p = 0.245$) and postpartum blood loss ($p = 0.526$).

Conclusion Third-trimester sFlt-1 serum level is not independently associated with PAS disorders or postpartum blood loss after adjusting for confounding factors.

Keywords Placenta accreta spectrum disorders · Postpartum blood loss · Pregnancy · Soluble fms-like tyrosine kinase 1 (sFlt1) · Third trimester · Enzyme · Linked immunosorbent assay (ELISA)

Abbreviations

sFlt-1 Soluble fms-like tyrosine kinase-1
PAS Placenta accreta spectrum
PA Placenta accreta
PI Placenta increta
PP Placenta percreta
VEGF Vascular endothelial growth factor

PIGF Placental growth factor
FIGO The international federation of gynecology and obstetrics
ELISA Enzyme-linked immunosorbent assay

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What does this study add to the clinical work

This study highlights the relationship between third-trimester maternal serum soluble fms-like tyrosine kinase-1 (sFlt-1) levels and placenta accreta spectrum (PAS) disorders, as well as postpartum blood loss. It provides valuable insights for early diagnosis and management of PAS disorders in clinical practice.

Introduction

Placenta accreta spectrum (PAS) disorders refer to a spectrum of conditions in which the chorionic villi abnormally invade the uterine myometrium to varying degrees.

Classification of PAS disorders depends on the extent of myometrial invasion as follows: placenta accreta (PA), placenta increta (PI), and placenta percreta (PP) [1, 2]. PAS disorders are life-threatening obstetrical complications associated with massive postpartum hemorrhage, shock, possible emergent hysterectomy as well as other serious surgical and postoperative complications [3–5]. A multicenter study in China showed that the incidence of the disease progressively increased from 0.18% in 2011 to 0.78% in 2015 [6]. According to a national survey conducted in the United States by the end of 2017, 1 in 313 women undergoing cesarean delivery were diagnosed with placenta accreta spectrum disorders [7]. Hence, it is important to provide precise prenatal diagnosis, which prepares the patient and the medical team for potential complications and otherwise unforeseen events during delivery and in the immediate postpartum period. Ultrasound is the preferred method for antepartum diagnosis of PAS disorders [8, 9]. However, a study in the UK reported that PAS disorders were suspected during antenatal ultrasound or MRI, in just 50% of the patients who were later found to have PAS disorders at the time of delivery [10]. Thus, the value of ultrasound for classifying PAS disorders and predicting postpartum bleeding is suboptimal. MRI may also be helpful in the detection of PAS disorders, but its accuracy is controversial. Several studies have attempted to explore certain serum markers for PAS disorders, in the hope of optimizing early diagnosis.

The understanding of the pathogenesis of PAS disorders, has led investigators to attempt to identify maternal serum biomarkers that could be used for the early detection of PAS disorders. Some serum markers have been recently introduced, such as vascular endothelial growth factor (VEGF), placental growth factor (PIGF) and some certain miRNAs [11–13]. Nevertheless, a sensitive serum biomarker for predicting PAS disorders has yet to be found [1]. Uterovesical hypervascularity is a prominent feature of PAS and may contribute to postpartum hemorrhage [8]. Concomitantly, soluble fms-like tyrosine kinase-1 (sFlt-1) is believed to exhibit an antiangiogenic effect [14]. Histological studies have demonstrated lower levels of sFlt-1 protein at the maternal–fetal interface in PAS disorders compared with normal placentation, from which it can be speculated that the lack of sFlt-1 may contribute to abnormal vascular reconstruction and the pathogenesis of PAS disorders by reducing its antiangiogenic effect [15, 16]. Maternal sFlt-1 level has been confirmed by some studies to be associated with preeclampsia, which is also a uteroplacental vascular-related disorder [17].

There are a few studies on the relationship between maternal circulating sFlt-1 level and PAS disorders, whose findings contradict each other. Until date, no study on sFlt-1 and postpartum blood loss has been conducted. The purpose of this research was to evaluate the circulating sFlt-1 level

in gravid women with or without PAS disorders in the third trimester and to elucidate the potential of maternal serum sFlt-1 level in predicting PAS disorders and postpartum bleeding during delivery.

Methods

Research object

Our study employed a nested case–control study design. Participants in this study were gravid women with a singleton gestation who delivered at the Peking University Third Hospital Obstetrics Department from September 2018 to January 2019. Women who met any of the following exclusion criteria were excluded: (1) incomplete clinical information; (2) inadequate volume of blood samples; (3) presence of concomitant diseases, including diabetes mellitus requiring insulin, rheumatic disease, preeclampsia or eclampsia. The inclusion criteria for the PAS group were antepartum suspicion of PAS disorders by ultrasound, which was confirmed by clinical diagnosis at delivery and/or postpartum histopathology. Other participants were divided in non-PAS group. The diagnosis of PAS disorders refer to the standard formulated by the International Federation of Gynecology and Obstetrics (FIGO) [2].

Study design

Peripheral blood and clinical data were collected. Baseline information were collected from the medical records, including maternal age at delivery, BMI before pregnancy, parity, number of previous cesarean deliveries, gestational age at blood taking and placental position. Clinical outcomes, such as gestational age at delivery, cesarean section, postpartum blood loss, postpartum blood transfusion volume, emergent surgical procedures including hysterectomy, transfer into ICU, birth weight of neonate, and transfer of neonate into NICU, were also collected.

Peripheral venous blood was collected during the third trimester of pregnancy. Venipuncture was used to obtain blood samples upon admission and before delivery. The samples were centrifuged at 3000 g at 4 °C for 15 min and frozen immediately at – 80 °C until the analyses were carried out. The sFlt-1 level in serum was measured using enzyme-linked immunosorbent assay (ELISA) kits (Boster Biological Technology Co. Ltd., Wuhan, Hubei, China) in accordance with the manufacturer's guidelines.

Serum sFlt-1 level was compared between the different groups. After that, to avoid bias from the impact of occlusion balloon placement in the abdominal aorta on blood loss, patients who received occlusion balloon placement in the abdominal aorta were excluded to explore the relationship

Table 1 Demographic characteristics, delivery outcomes, and neonatal outcomes in individuals in the PAS group and non-PAS groups

	PAS Group (<i>n</i> = 36)	Non-PAS Group (<i>n</i> = 24)	P value
Demographic characteristics			
Maternal age (y)	33.5 ± 3.9	35.8 ± 3.4	0.020
BMI before pregnancy (kg/m ²)	23.2 ± 2.4	22.8 ± 2.5	0.463
Parity	1 [0, 1]	1 [0.3, 1]	0.918
Previous Cesarean section	21 (58.3)	15 (62.5)	0.747
Gestational age at initial blood taking (weeks)	35.6 [33.1, 36.7]	38.6 [37.1, 39.0]	<0.001
Delivery outcomes			
Cesarean section	35 (97.2)	23 (95.8)	0.644
Placenta previa	33 (91.7)	8 (33.3)	<0.001
Gestational age at delivery (weeks)	36.6 [34.9, 37.3]	38.7 [37.5, 39.0]	<0.001
Postpartum blood loss (ml)	1400 [650, 2000]	200 [200, 375]	<0.001
Postpartum blood transfusion volume (U)	4 [0, 11]	0 [0, 0]	<0.001
Neonatal outcomes			
Birth weight (g)	2815.0 [2352.5, 3192.5]	3310.0 [2982.5, 3700.0]	<0.001
Transferred to the NICU	10 (27.8)	1 (4.20)	0.048

Continuous were expressed as means and standard deviations ($\bar{x} \pm s$) and assessed using an independent two-sample t-test. The Mann–Whitney U test was used to analyze continuous data that did not follow the normal distribution. The data were expressed as medians and interquartile ranges (M[P25, P75]). The Chi-Square test was used to compare categorical data, reported as N (%)

between serum sFlt-1 level and postpartum blood loss in patients with PAS disorders. All gravid women signed informed consent forms after receiving complete information about the study. The Research Ethics Committee of Peking University Third Hospital approved the protocol (approval S2018205).

Statistical analysis

Statistical analysis was performed using SPSS 22.0 for windows. Continuous data, conforming to the normal distribution through the Shapiro–Wilk test, were expressed as means and standard deviations ($\bar{x} \pm s$) and assessed using an independent two-sample t-test. The Mann–Whitney U test was used to analyze continuous data that did not follow the normal distribution. The data were expressed as medians and interquartile ranges (M[P25, P75]). The Chi-Square test was used to compare categorical data, reported as N (%). Spearman correlation analysis and Receiver Maternal age and gestational age at blood taking were adjusted. All statistical tests were two-sided, with a *p* value < 0.05 deemed statistically significant.

Results

Baseline characteristics of subjects

Thirty-eight cases met the inclusion criteria for the PAS group but two cases were excluded due to insulin use.

Thirty-six cases of PAS disorders were recruited, including 13 cases of placenta accreta, 13 cases of placenta increta, and 10 cases of placenta percreta. Twenty-four cases were recruited in the non-PAS group, including eight cases of women with placenta previa and 16 cases of women with normal placentation.

Compared with the non-PAS group, women with PAS disorders had higher postpartum blood loss volumes, higher postpartum blood transfusion volumes, and earlier gestational ages at delivery. The proportion of placenta previa was higher in the PAS group. The gestational age at blood taking in the PAS group (35.6 weeks) was significantly earlier than that in the non-PAS group (38.6 weeks) (*p* < 0.001). In the PAS group, five women required an emergent hysterectomy, and eight women required transfer to the ICU. The infants of women with PAS disorders had lower birth weights than those of the non-PAS group. No asphyxia was reported in any of the newborns (Table 1).

Comparisons of serum sFlt-1 levels in different groups

The sFlt-1 median level in women with PAS disorders was 9407.1 (2745.9–21,691.5) pg/mL, and was 25,779.2 (14,317.1–35,626.7) in the non-PAS group. (*P* < 0.001, Fig. 1). Decreasing sFlt-1 level was observed with increasing severity of PAS disorders. The sFlt-1 level in the PA, PI, and PP groups were 11,757.2 (7429.8–22,263.5) pg/mL, 5271.0 (2380.3–24,446.7) pg/mL, and 2416.3 (1536.6–16,237.6)

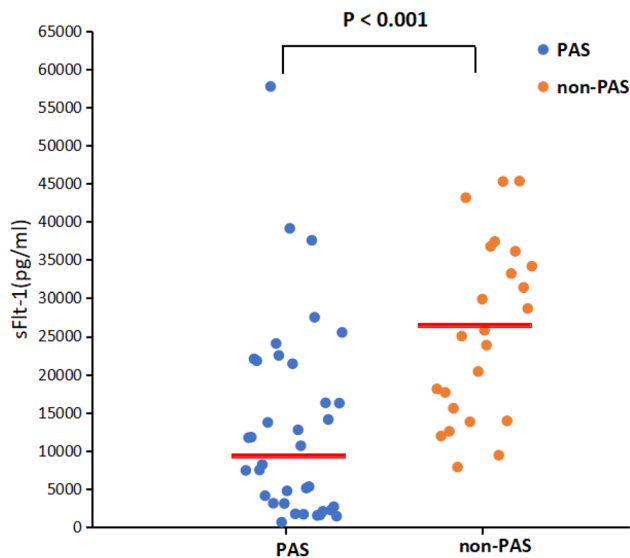


Fig. 1 Maternal serum sFlt-1 level in the PAS group and non-PAS group

pg/mL, respectively ($p=0.158$) (Fig. 2). There were significant differences in the sFlt-1 level between the non-PAS subgroups ($P=0.041$). Serum sFlt-1 level in the placenta previa was $18,967.8 (\pm 11,023.8)$ pg/mL and in the normal placentation group was $29,122.1 (\pm 10,692.3)$ pg/mL (Fig. 2).

In this research, there was a difference in gestational age at blood taking between PAS group and non-PAS group, and sFlt-1 level was associated with gestational age at blood sampling. As gestational age at blood taking increased, sFlt-1 level showed an upward trend (Fig. 3) ($r=0.654$, $p<0.001$). Therefore, maternal age and gestational age at blood taking were adjusted. The result showed that sFlt-1

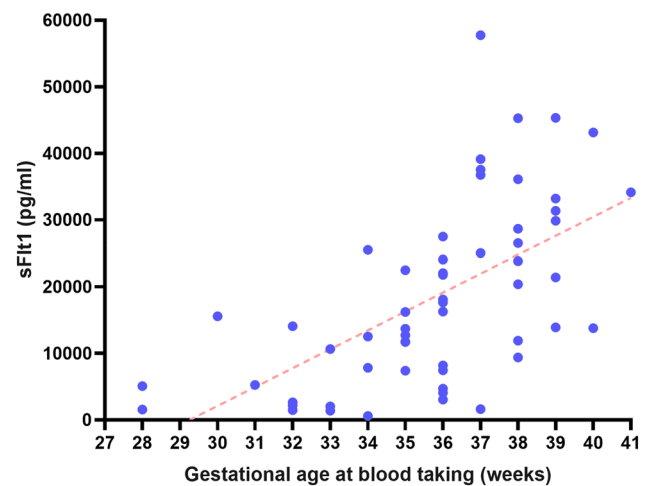


Fig. 3 Trend of sFlt-1 level with gestational age at blood sampling. The red line represents the trend of maternal serum sFlt-1 level with gestational age at blood taking ($r=0.654$, $p<0.001$)

level was not significantly associated with PAS disorders (Table 2) ($p=0.245$).

Relationship between sFlt-1 and postpartum blood loss

To avoid bias from the impact of occlusion balloon placement in the abdominal aorta on blood loss, three patients were excluded, leaving 33 cases in the PAS group (13 with placenta accreta, 12 with placenta increta, and eight with placenta percreta). The maternal sFlt-1 level was negatively correlated with the severity of postpartum blood loss (Fig. 4) ($r=-0.358$, $p=0.041$).

After adjusting for maternal age and gestational age at blood taking, the partial correlation analysis revealed no

Fig. 2 Serum sFlt-1 level in women with placenta percreta (PP), placenta increta (PI), placenta accreta (PA), placenta previa, and normal placentation. Red lines mark group medians

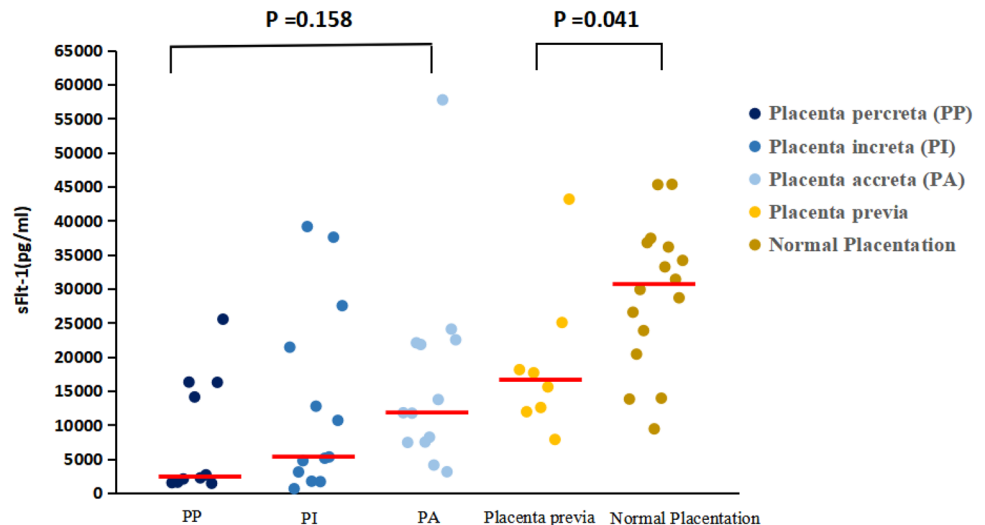


Table 2 Adjusted relationship between sFlt-1 level and the occurrence of PAS

Variable	SE	p-value	OR	95% CI for OR
Gestational age at initial blood taking	0.244	0.008	0.522	(0.324, 0.842)
sFlt-1 level	0.000	0.245	1.000	(1.000, 1.000)
Maternal age	0.098	0.041	0.819	(0.677, 0.992)

SE standard error, OR odds ratio, CI confidence interval

significant association between sFlt-1 level and postpartum blood loss (Table 3) ($r = -0.118$, $p = 0.526$).

Discussion

Serum markers for PAS disorders have been investigated to improve the diagnostic accuracy and predict disease severity. This study investigated the serum sFlt-1 level of gravid women in third trimester in relation to PAS disorders and postpartum bleeding. Serum sFlt-1 level in the third trimester was decreased in patients with PAS disorders and related with postpartum blood loss. However, after adjusting for confounding factors (maternal age and gestational age at blood taking), the study overall presented negative results.

Considering that serum sFlt-1 level is related to gestational age [18], The results of this study are contrary to the expected outcomes. As a soluble vascular endothelial growth factor (VEGF) and PIGF receptor [19], sFlt-1 can combine

Table 3 Partial correlation analysis of sFlt-1 with postpartum blood loss

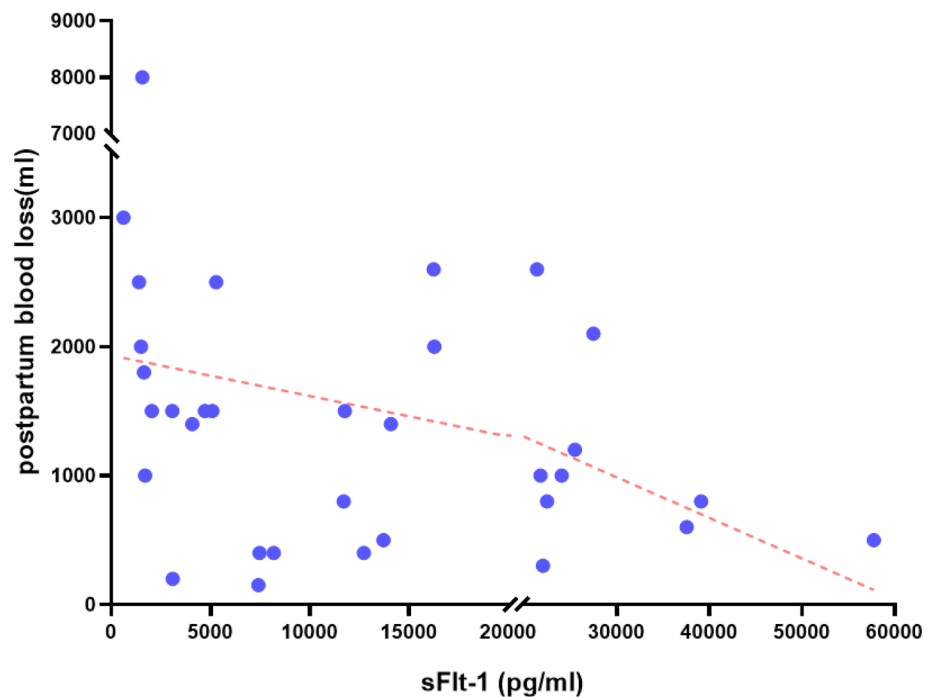
Variable	r	p-value
sFlt-1 level	-0.118	0.526

Maternal age and gestational age at blood taking were adjusted. r: partial correlation coefficient

with these ligands and inhibit their angiogenesis functions, thus inhibiting angiogenesis [20]. A previous study reported that sFlt-1 level in women with placenta percreta at 34.59 weeks was 236.28 ± 48.44 ng/ml, which was significantly lower than the levels in women with normal placentation at 35.36 weeks of gestation (276.30 ± 77.72 ng/ml, $p < 0.05$) [21]. We hypothesized that decreased serum sFlt-1 levels in patients with PAS disorders may be associated with the pathophysiological changes in the placenta. Similar to our study results, Biberoglu et al. [22] reported that sFlt-1, PIGF, sFlt-1/PIGF ratio, and VEGF in maternal serum did not show a significant difference between women with placenta previa, placenta accreta/increta, placenta percreta, and normal placentation women in the control group. The gestational age at blood taking were 35.7 weeks, 35.2 weeks, 35.9 weeks, and 38.4 weeks, respectively. Studies with larger sample sizes are needed to verify the predictive value of sFlt-1 for postpartum blood loss. Besides, the estimation of postpartum blood loss is somewhat subjective, which may affect the study results.

This research has several limitations that preclude us from drawing definitive conclusions. First, this study was

Fig. 4 The relationship between sFlt-1 level and postpartum blood loss volume. The red line represents the trend of postpartum blood loss volume with sFlt-1 level in the PAS group using linear regression ($r = -0.358$, $p = 0.041$)



conducted at a single research center, and it encountered the challenge of a limited sample size due to the low incidence rate of PAS disorders. Second, due to the difference in gestational age at blood taking between the two groups, it is difficult to avoid its impact on sFlt-1 levels. Future studies will include a larger sample size for matched analysis.

Author contributions Fangchao Zhang: statistical analysis, manuscript writing. Li Xia: data collection, study designing, manuscript writing. Lin Zeng: data analysis. Huanyu You: data collection. Qingao Liu: manuscript collating. Yan Wang: project development.

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Data availability The data that support the findings of this study are available on request from the corresponding author, upon reasonable request.

Declarations

Conflict of interest The authors have no relevant financial or non-financial interests to disclose.

Ethical approval This study was performed in line with the principles of the Declaration of Helsinki. The Research Ethics Committee of Peking University Third Hospital approved the protocol (approval S2018205). All methods were performed in accordance with the relevant guidelines and regulations.

Consent to participate Informed consent was obtained from all individual participants included in the study.

Consent to publish All participants signed informed consent forms and approved the publication of this article.

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